

Source Integration

- NCERT Class 7 — Our Environment, Chapter 7: Human Environment – Settlement, Transport and Communication

Core Factual & Conceptual Architecture

1. Evolution of Human Settlements

- Definitions: Settlements are places where people construct homes. Early humans transitioned from trees and caves to permanent structures once agriculture developed, clustering near river valleys like the Indus, Tigris, Nile, and Hwang-He.
- Site Factors: The specific site selection depends on favorable climate, water availability, suitable land, and fertile soil.
- Settlement Typologies:
 - Temporary Settlements: Occupied briefly by nomadic or hunting populations, including pastoralists practicing **transhumance** (seasonal movement between pastures).
 - Permanent Settlements: Long-term housing formats that dominate modern human geography.

2. Rural versus Urban Morphology

- Rural Settlements: Engaged in primary activities like agriculture, forestry, and fishing. Patterns include **compact settlements** (closely built dwellings on flat land) and **scattered settlements** (spread out in hilly tracts or extreme climates). Architecture adapts to local climates (e.g., slanting roofs in heavy rainfall zones, stilts for flood-prone areas).
- Urban Settlements: Towns and cities driven by secondary and tertiary activities (manufacturing, trade, services). Rapid rural-to-urban migration generates urban problems such as traffic congestion, pollution, slums, and utility shortages.

3. Transport Infrastructure

- Roadways: Crucial for short distances; categorized into metalled (paved) and unmetalled (unpaved) roads. Notable infrastructure includes the Manali-Leh highway and India's Golden Quadrilateral.
- Railways: Efficient for heavy freight and long-distance passenger transit. Key benchmarks include India's network (largest in Asia) and the Trans-Siberian Railway (world's longest, linking St. Petersburg to Vladivostok).
- Waterways: Cheapest transport mode for bulky goods, divided into inland waterways (Ganga-Brahmaputra, Great Lakes, Nile) and oceanic sea routes connecting global ports.
- Airways: Fastest and most expensive transport mode, uniquely capable of reaching remote regions and executing disaster relief via helicopters.

4. Modern Communication Networks

- Evolution: Driven by technological innovation into an **information revolution**.
- Mass Media: Newspapers, radio, and television broadcasting to large audiences simultaneously.
- Advanced Technology: Satellites facilitate weather forecasting, resource surveys, and global positioning, while the internet, e-mail, and cellular phones enable instantaneous personal and commercial interaction, making the world feel like a global village.